CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents what the MedShield Decision Support System currently demonstrates and what the available evidence does not yet support. The strongest result is a traceable descriptive layer connected to a working dashboard: 37,728 accepted sales facts from 2017 to 2025 reconcile to Databricks Gold candidate views, missing months remain visible, and the interface can compare sales, products, areas, and transparent forecast baselines. Predictive model publication, external-signal forecasting, and operational inventory optimization remain gated. We therefore discuss them as evaluated or implemented capabilities, not as proven business outcomes.

The discussion follows the project objectives and the North Star question: How can MedShield use historical sales data, validated external signals, and scenario-based planning to improve demand visibility and reduce expiry-driven inventory losses for the 2026 planning cycle? Each result is labeled according to its evidence status. Observed sales are reported as historical facts; mapped labels and ABC classes are candidates; forecast comparisons are evaluation evidence; weather data from NASA POWER is a proxy; and planning outputs are scenarios that require human review. This treatment keeps the chapter aligned with the implemented system and prevents unfinished work from being presented as a completed operational result.

4.1 Presentation of Results

4.1.1 Alignment With the Research Questions and Objectives

The proposal defined five objectives: describe sales and seasonality, prioritize products and areas, forecast demand with external signals, support inventory decisions, and present the results through five dashboard modules. The implementation moved furthest on the first, second, and fifth objectives. Forecasting has reached controlled evaluation but not release approval. The prescriptive layer can solve constrained scenarios, but the study does not contain the stock, expiry, lead-time, supplier, ordering-cost, or holding-cost data needed to measure an actual reduction in wastage.

Table 4.1 Objective achievement audit

Objective	Evidence expected	Observed result	Assessment
SO1	Sales performance, year-over-year patterns, and STL seasonality	Historical summaries and a 72-month STL decomposition are available. Ten earlier months remain unobserved, and the full geography mapping still needs owner approval.	Partially achieved
SO2	Product, territory, and account priorities using Pareto ABC and predictive scoring	Deterministic Pareto and ABC candidate views are implemented. Product labels are not approved SKUs, and no XGBoost classifier is published.	Partially achieved
SO3	Territory forecasts using sales, DII, RSI, Prophet, and XGBoost	Sales-only baselines and offline candidates were evaluated. External joins have zero forecast-eligible rows, and no model is approved or activated.	Not achieved for operational use
SO4	EOQ, ROP, and allocation decisions that reduce expiry wastage to at most 5 percent	The interface supports reviewed allocation scenarios. Real inventory and expiry outcomes are absent, so the 5 percent target was not tested.	Scenario capability only
SO5	Five dashboard modules with upload validation and decision support	The five modules and gateway-backed descriptive workflow are implemented. Live Signal 2 alerts, collaborative filtering, and automated procurement are not verified outputs.	Substantially achieved with exclusions

Source: Capstone proposal objectives and the MedShield current-state, analytics, and QA records as of 27 September 2026.

This audit is important for defense because completion of software features is not the same as achievement of a research outcome. A screen can exist even when its model is not approved, and a solver can run even when its inputs are hypothetical. The remainder of the chapter reports both the implemented function and the evidence boundary that applies to it.

4.1.2 Implemented System and Runtime Architecture

MedShield was implemented as an authenticated web application. The Next.js frontend displays analytical views and sends requests to an Express and TypeScript gateway. The gateway performs validation, authorization, and Databricks query orchestration. Databricks Unity Catalog stores the analytical source of truth through Bronze, Silver, Audit, and Gold candidate schemas. Supabase supplies identity and audit support; it does not provide the dashboard measures. Python services are used for offline preparation, model experiments, and validation, but they are not a browser fallback.

The runtime boundary was deliberately kept strict. If Databricks or a required view is unavailable, the gateway returns an error and the dashboard shows an unavailable state. It does not replace the missing result with local JSON, seeded values, or an old spreadsheet. This behavior improves traceability because every displayed analytical value has one declared runtime source.

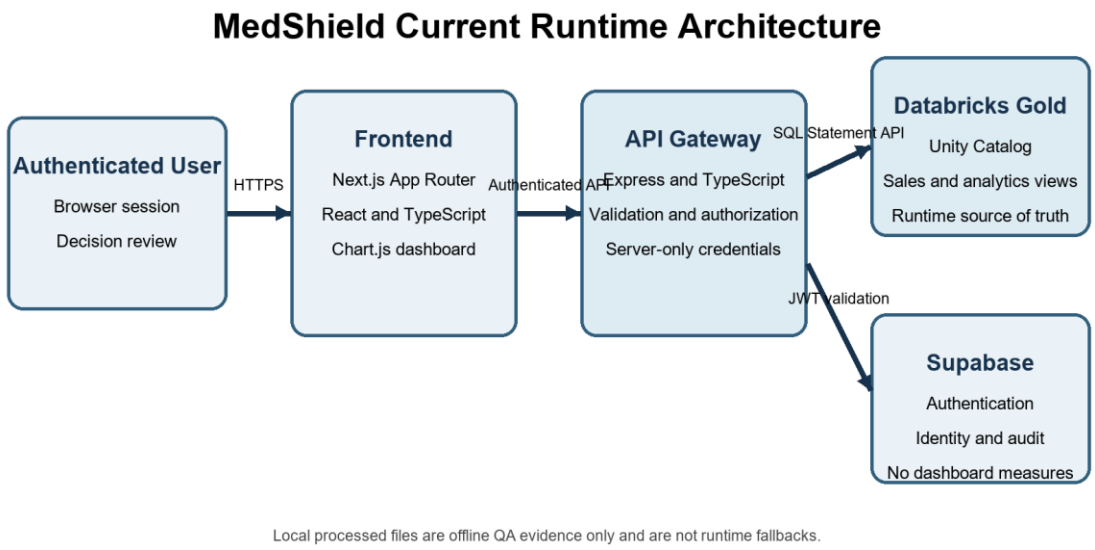


Figure 4.1 MedShield runtime architecture

Source: MedShield Architecture and Database documentation, September 2026.

Table 4.2 Implemented system layers

Layer	Implemented component	Responsibility	Boundary
Presentation	Next.js, React, TypeScript, Chart.js	Filters, charts, tables, exports, readiness labels, and scenario forms	Does not query Databricks or external providers directly
Gateway	Express and TypeScript	Authentication checks, parameter validation, query orchestration, and response contracts	Server-only credentials; explicit unavailable states
Analytical store	Databricks Unity Catalog	Bronze source evidence, Silver assessed records, Gold candidate facts and aggregates	Candidate publication pending business approval
Identity and audit	Supabase Auth and audit persistence	Session identity, roles, and application audit events	Not an analytical source
Offline analytics	Python jobs and tests	Reconciliation, STL, model evaluation, and scenario validation	Not a live dashboard dependency

Source: MedShield Architecture, Database, Security, and Project documentation.

4.1.3 Data Sources Coverage and Quality

The historical sales candidate covers 2017 to 2025 and contains 37,728 accepted fact rows. The study proposal emphasized 2021 to 2025 and three Philippine regions, but the implemented warehouse retains the longer history available in the supplied records. We used the broader period for context while keeping the proposed regional interpretation conditional on the approved area mapping. Department of Health and weather sources are retained as historical context; they are not silently joined to sales for forecast publication.

Table 4.3 Data sources and current analytical status

Source	Coverage	Current use	Status and limitation
MedShield sales	2017 to 2025	Historical demand, revenue, product, area, and baseline forecast evaluation	37,728 accepted facts; candidate publication
DOH surveillance	2018 to 2026 declared; 2026 partial	Historical disease context and candidate regressors	Territory join not approved; no individual health inference
PAGASA observations	2017 to 2024	Official historical weather context	No official 2025 observations in the project; no approved sales match
NASA POWER	108 monthly rows for nine areas in 2025	Separately labeled weather proxy	Provider proxy, not PAGASA; 32 candidate matches do not approve regression
Operational inventory	Not supplied	Would support EOQ, ROP, safety stock, allocation, and expiry analysis	Scenario inputs only

Source: MedShield Current State, System Documentation, and publication-readiness evidence.

The quality review identified a complete 108-month calendar but only 98 months with observed transactions. The ten unavailable months are May, June, July, August, and October 2017; January and February 2018; and October, November, and December 2019. These are missing observations, not zero demand. The system preserves them as null periods so that trend and validation routines do not create artificial declines.

Table 4.4 Sales candidate quality profile

Quality measure	Observed value	Interpretation
Accepted fact rows	37,728	Checksum-bound candidate population
Revenue-eligible rows	36,848	Rows included in the net-sales measure
Margin-eligible rows	34,105	Rows with a valid margin denominator
Warning-bearing rows	32,749	Warnings remain visible and are not equivalent to rejection
Calendar coverage	108 months total; 98 observed; 10 missing	Missing periods remain unavailable rather than zero-filled
Product identity	4,561 source labels; 0 approved master entries	Product findings describe labels, not verified canonical SKUs
Area identity	22 derived labels	Some labels mix geography, customer type, or business line and require approval
Returns evidence	673 rows	Signed returns remain separate while contributing to net revenue

Source: Version-bound publication and QA evidence for the 37,728-row Databricks candidate.

4.2 Data Analysis

4.2.1 Historical Sales Performance and Seasonality

Across the accepted candidate, observed net sales totaled ₱626,946,741.81. Workbook gross profit totaled ₱288,487,498.72. The weighted gross-margin rate was 60.26 percent, calculated only over the ₱478,720,670.03 margin-eligible revenue basis. This is a transaction-level workbook measure and should not be described as company net income. The separate denominators also explain why dividing gross profit by all reported net sales would give the wrong percentage.

Table 4.5 Annual sales and workbook gross profit

Year	Transactions	Net sales	Workbook gross profit	Margin eligible sales
2017	1,978	₱10,211,167.35	₱6,316,157.68	₱9,212,606.10
2018	6,181	₱51,619,691.27	₱28,328,721.76	₱44,882,495.39
2019	5,446	₱62,542,304.49	₱29,125,469.95	₱41,211,281.42
2020	3,980	₱63,593,274.63	₱28,013,993.68	₱40,259,988.09
2021	3,407	₱77,860,206.26	₱50,420,770.93	₱77,804,247.26
2022	3,980	₱63,234,990.04	₱40,799,049.06	₱60,493,221.21
2023	5,381	₱62,228,966.85	₱33,585,582.48	₱59,822,843.64
2024	2,831	₱50,259,319.88	₱19,563,837.42	₱49,956,893.05
2025	4,544	₱185,396,821.04	₱52,333,915.76	₱95,077,093.87

Source: Descriptive calculation receipt. Amounts are candidate values in Philippine pesos.

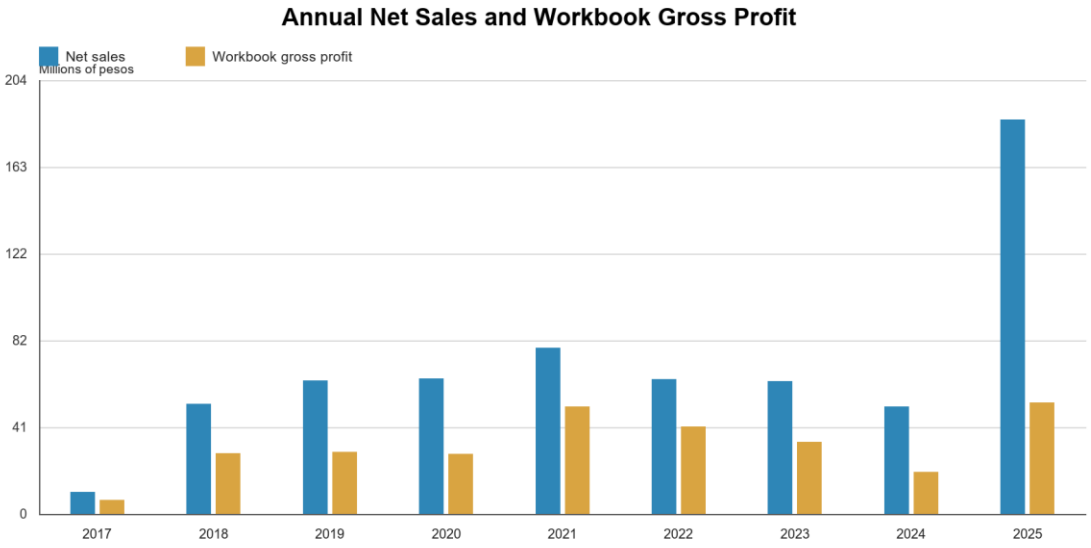


Figure 4.2 Annual net sales and workbook gross profit

Source: Generated from the version-bound descriptive calculation receipt.

The annual series should be interpreted as observed recorded activity, not as proof that every year is equally complete. The 2025 revenue increase is material in the candidate dataset, but the year also contains source adjustments whose product attribution remains under review. For this reason, we can defend the recorded financial total while keeping product-level and causal explanations provisional.

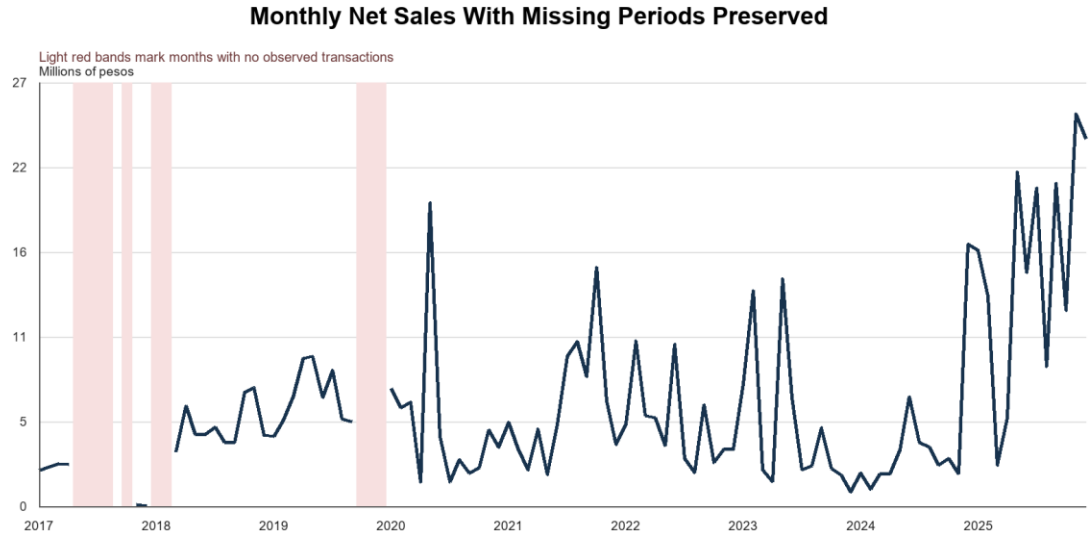


Figure 4.3 Monthly net sales and missing coverage

Source: Generated from the 108-month Databricks descriptive snapshot.

Observed calendar-month averages were highest in May at ₱9.87 million and June at ₱7.55 million. This result indicates a mid-year pattern in the recorded series, but it is not the same as a statistical seasonal component. The dashboard therefore keeps the calendar-month average separate from the STL decomposition required by the first objective.

Table 4.6 Observed calendar month averages

Month	Average net sales	Observed years
Jan	₱6,405,176.08	8
Feb	₱7,098,050.31	8
Mar	₱3,879,490.79	9
Apr	₱4,433,381.20	9
May	₱9,871,031.23	8
Jun	₱7,553,552.93	8
Jul	₱6,832,765.69	8
Aug	₱5,074,209.17	8
Sep	₱6,073,372.18	9
Oct	₱6,538,044.40	7
Nov	₱6,502,031.32	8
Dec	₱7,126,212.47	8

Source: Descriptive snapshot. Method is observed calendar-month average, not STL.

The STL routine used a 12-month period, a seasonal window of 13, robust fitting, and no imputation. Because the early history contains gaps and short segments, only the contiguous January 2020 to December 2025 series qualified for decomposition. The resulting 72 months reconstruct the observed series to numerical precision. The decomposition describes the structure of this candidate series; it does not establish that disease or weather caused the seasonal movement.

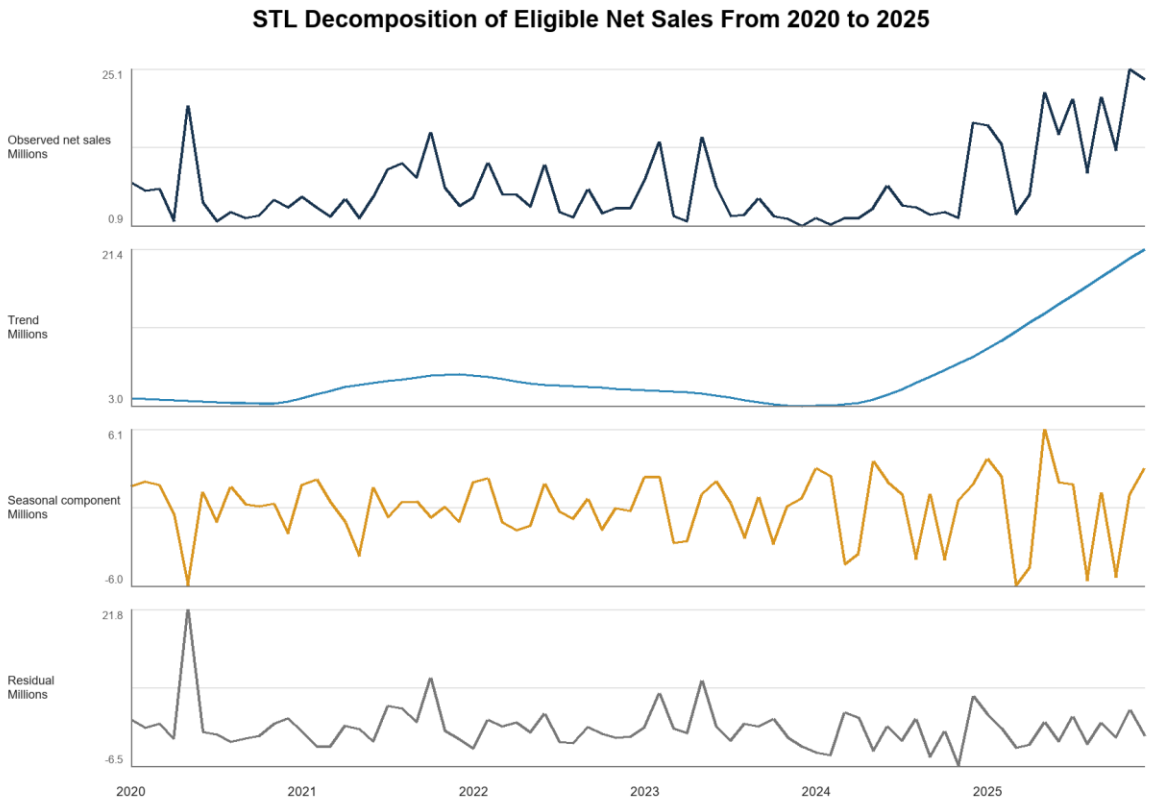


Figure 4.4 STL decomposition of eligible net sales

Source: Checksum-bound STL evidence generated with statsmodels 0.14.6.

4.2.2 Product and Area Prioritization

The second objective asks which products, territories, and accounts should receive attention. The implemented descriptive layer answers this through positive-sales contribution and deterministic Pareto or ABC grouping. It does not use an approved predictive XGBoost classifier. The distinction matters because an ABC class describes contribution within the selected history; it does not predict future urgency or stock-out risk.

Table 4.7 Top source product labels by net sales

Rank	Source product label	Net sales	Share	ABC
1	SPEEDA 2.5IU/0.5ML	₱22,466,448.23	3.57%	A
2	PROHAB 10MG/ML 20ML	₱16,530,532.11	2.63%	A
3	BUPIRIGHT 5MG/ML 0.5% IN 8% 4ML	₱15,966,455.39	2.54%	A
4	SITIXON 1G	₱13,051,271.68	2.08%	A
5	MONOWEL 1G IV	₱12,313,552.50	1.96%	A
6	PROFUREX 750MG	₱11,987,494.40	1.91%	A

Rank	Source product label	Net sales	Share	ABC
7	EVAPROST 250MCG/ML	₱9,753,530.12	1.55%	A
8	F6 FETAL MONITOR SINGLE	₱7,848,998.50	1.25%	A
9	SPEEDA	₱7,558,970.97	1.20%	A
10	EUROXONE 1G	₱7,311,190.91	1.16%	A

Source: Databricks product candidate view. Labels are not approved canonical SKUs.

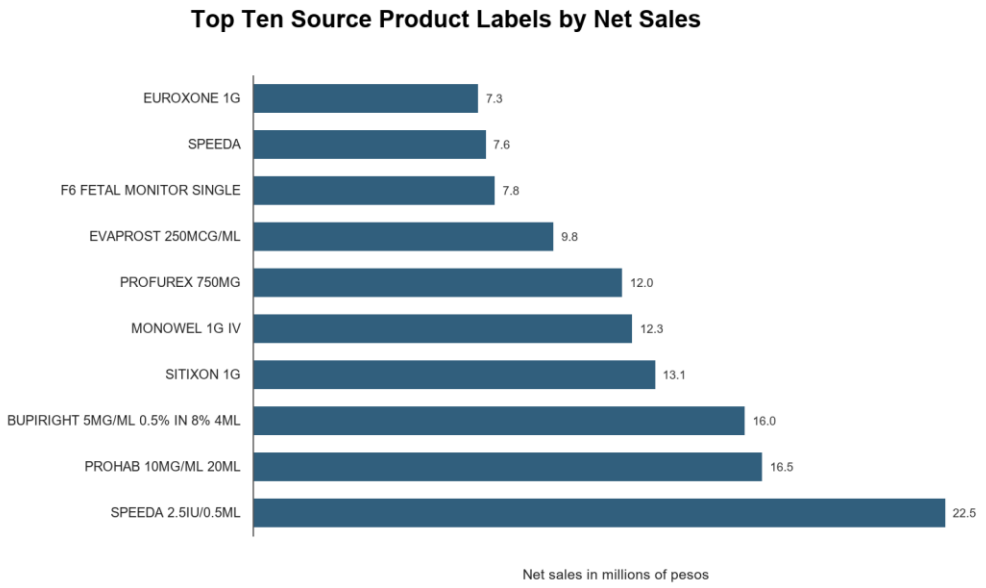


Figure 4.5 Top ten source product labels by net sales

Source: Generated from the descriptive calculation receipt.

SPEEDA 2.5IU/0.5ML ranked first with ₱22,466,448.23, or 3.57 percent of total candidate net sales. The top five labels contributed 12.78 percent, and the top ten contributed 19.85 percent. These figures do not support a claim that a handful of products dominate the entire portfolio. They instead show a long-tail distribution in which prioritization needs a full-population Pareto calculation. Alias fragmentation may also split one commercial product across several labels, as illustrated by the separate SPEEDA entries.

Table 4.8 Top derived area labels by net sales

Rank	Derived label	Net sales	Share	Workbook gross profit
1	GOVERNMENT	₱310,629,101.40	49.55%	₱130,353,976.35
2	HOSPITAL	₱165,121,239.71	26.34%	₱69,103,315.46
3	BATANGAS	₱28,556,256.25	4.55%	₱17,838,380.41
4	QUEZON	₱27,207,048.54	4.34%	₱17,340,506.74

Rank	Derived label	Net sales	Share	Workbook gross profit
5	ADMIN	₱16,359,733.69	2.61%	₱8,706,600.10
6	EQUIPMENT	₱14,475,259.73	2.31%	₱7,679,164.32
7	LAGUNA	₱14,376,324.92	2.29%	₱8,509,063.56
8	MARINDUQUE	₱13,469,314.65	2.15%	₱6,984,765.63
9	CAM NORTE	₱11,638,151.07	1.86%	₱7,508,275.57
10	CAVITE	₱7,799,599.04	1.24%	₱5,155,855.62

Source: Databricks area candidate view. Mixed commercial labels; not a pure geography dimension.

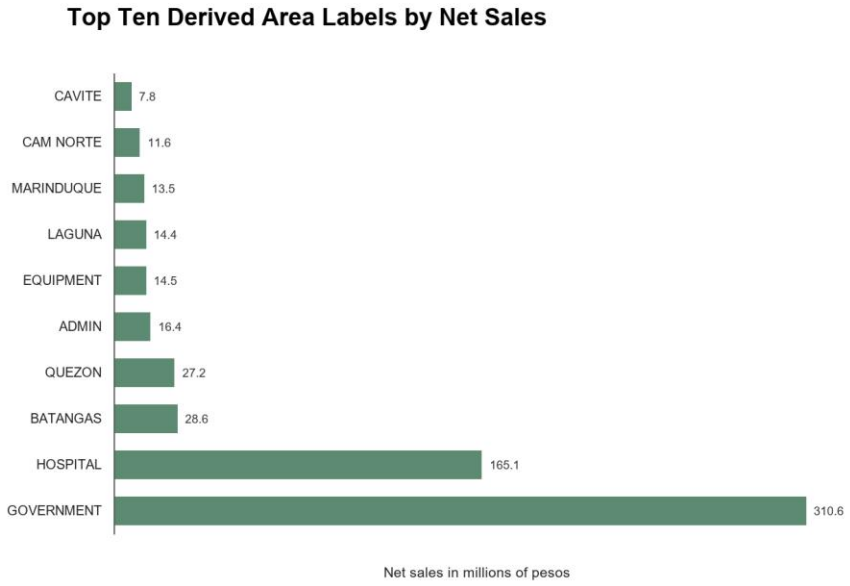


Figure 4.6 Top ten derived area labels by net sales

Source: Generated from the descriptive calculation receipt.

GOVERNMENT contributed 49.55 percent of candidate net sales and HOSPITAL contributed 26.34 percent. These labels are useful commercial segments, but they are not geographic territories. Batangas, Quezon, Laguna, Marinduque, Camarines Norte, and Cavite appear in the same legacy area field. The dashboard now exposes buyer-sector and geographic mappings separately, but final owner approval remains open. A defensible interpretation is that government-labeled transactions are financially important in the current candidate; it would be too strong to call that label a verified territorial market share.

4.2.3 Forecast Evaluation and External Signals

The third objective was only partly realized. The system can evaluate simple sales-only baselines and persist candidate experiment evidence, but it has no approved operational forecast.

In the fixed-origin Government net-sales comparison retained by the dashboard evidence, the last-observed-value method outperformed the seasonal-naive benchmark across the same nine observed holdout months. Both errors were high, and both methods underpredicted on average. The result supports transparent baseline comparison, not model promotion.

Table 4.9 Fixed origin Government net sales benchmark

Method	Paired months	MAE	RMSE	WAPE	Bias
Seasonal naive	9	₱2,695,359.26	₱3,459,768.58	85.05%	-₱2,574,060.06
Last observed value	9	₱1,987,359.74	₱2,632,312.49	62.71%	-₱1,230,712.16

Source: Section 8 historical acceptance evidence. Missing actuals were excluded rather than set to zero.

A later offline experiment evaluated 15 configurations: four simple benchmarks, two ETS models, and nine Prophet configurations. Last value remained the development-selected candidate, with 83.20 percent development WAPE and 37.76 percent retrospective 2025 WAPE. No challenger met the promotion criteria, empirical uncertainty intervals were unsupported, and the model release pointer remained empty. These results are stronger evidence than reporting a model merely because it ran. They show that the current forecast is not accurate or complete enough for an operational procurement decision.

Table 4.10 Predictive readiness audit

Evidence item	Observed result	Interpretation
Candidate configurations	15 evaluated	Experiment completed and persisted as candidate evidence
Development selection	Last value; WAPE 83.20%	Selected among tested candidates but high error
Retrospective 2025 audit	Last value; WAPE 37.76%	Historical audit only; not a 2026 holdout
Intervals	Unsupported	Insufficient lead-specific error history for defensible bands
DOH and PAGASA forecast rows	0 forecast eligible	External regression remains blocked
Publication	No active model release	No operational forecast may be claimed

Source: MedShield Analytics, Predictive Operations, Current State, and QA records dated 27 September 2026.

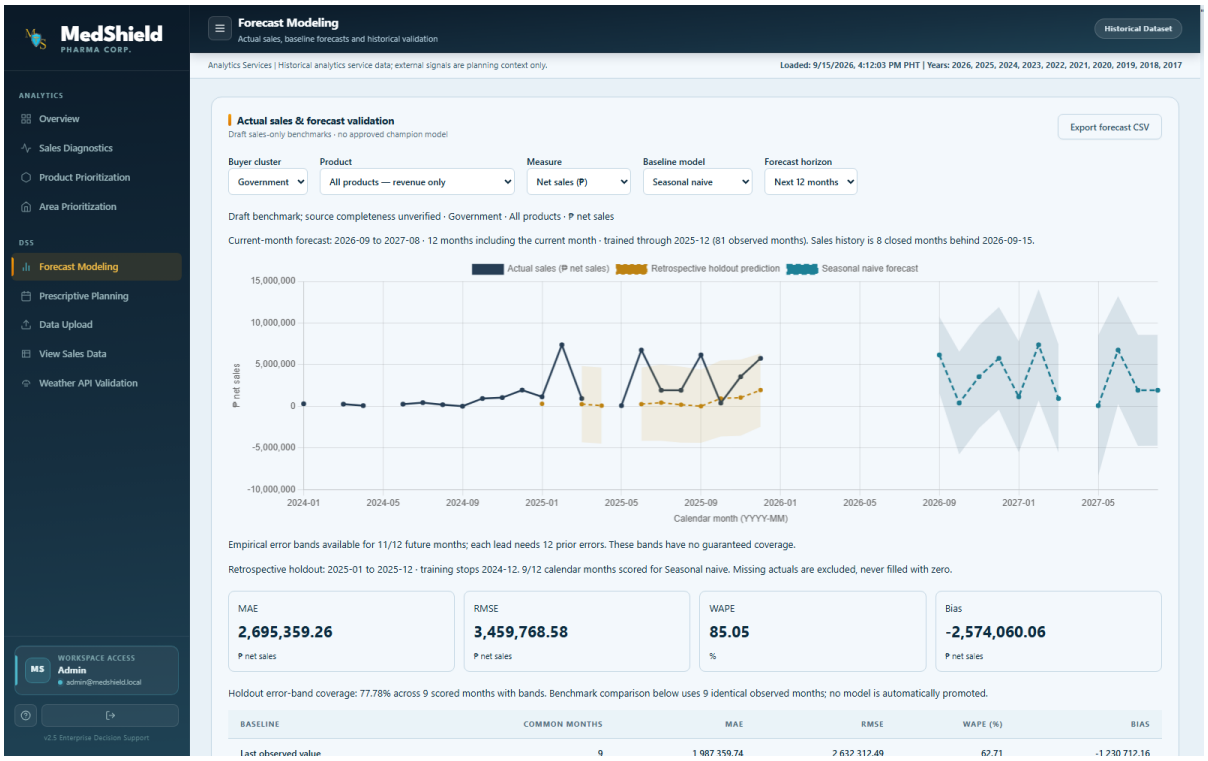


Figure 4.7 Forecast validation interface

Source: MedShield development interface capture dated 15 September 2026. The capture demonstrates the benchmark workflow and does not represent model publication approval.

The external data audit reinforces this boundary. The current candidate contains 2,171 DOH and PAGASA rows, but none are join-ready or forecast-eligible. NASA POWER supplies separately labeled proxy observations and 32 candidate sales matches; those matches do not establish an approved geographic join or predictive benefit. We therefore cannot claim that rainfall, disease intensity, or typhoon signals improved MedShield demand forecasts. Older exploratory files that label weather or disease models as validated are superseded by the current source-identity and publication gates.

4.2.4 Prescriptive Planning Results

The fourth objective is represented in the system as a controlled planning scenario. The interface can shortlist positive-revenue products, collect user-entered demand, usable stock, protected stock, pack size, cost, supplier limits, and budget, and solve a constrained allocation. Tests cover feasibility, integer packs, stock protection, and a service-first objective. The result is useful for demonstrating how a decision can be structured, but it remains conditional on the values entered by the user.

Table 4.11 Operational inputs required for prescriptive use

Input	Current status	Why it matters
On-hand and reserved stock	Not published	Determines actual availability and stock gaps
Expiry dates and batch age	Not published	Required to measure expiry-driven wastage
Lead time and service level	Not published	Required for ROP and safety-stock calculations
Ordering and holding cost	Not published	Required for defensible EOQ
Pack size MOQ and supplier cap	User-entered assumption	Constrains feasible orders
Budget and approved demand	User-entered assumption	Determines the allocation objective and ceiling

Source: MedShield business definitions, analytics guidance, and planning interface.

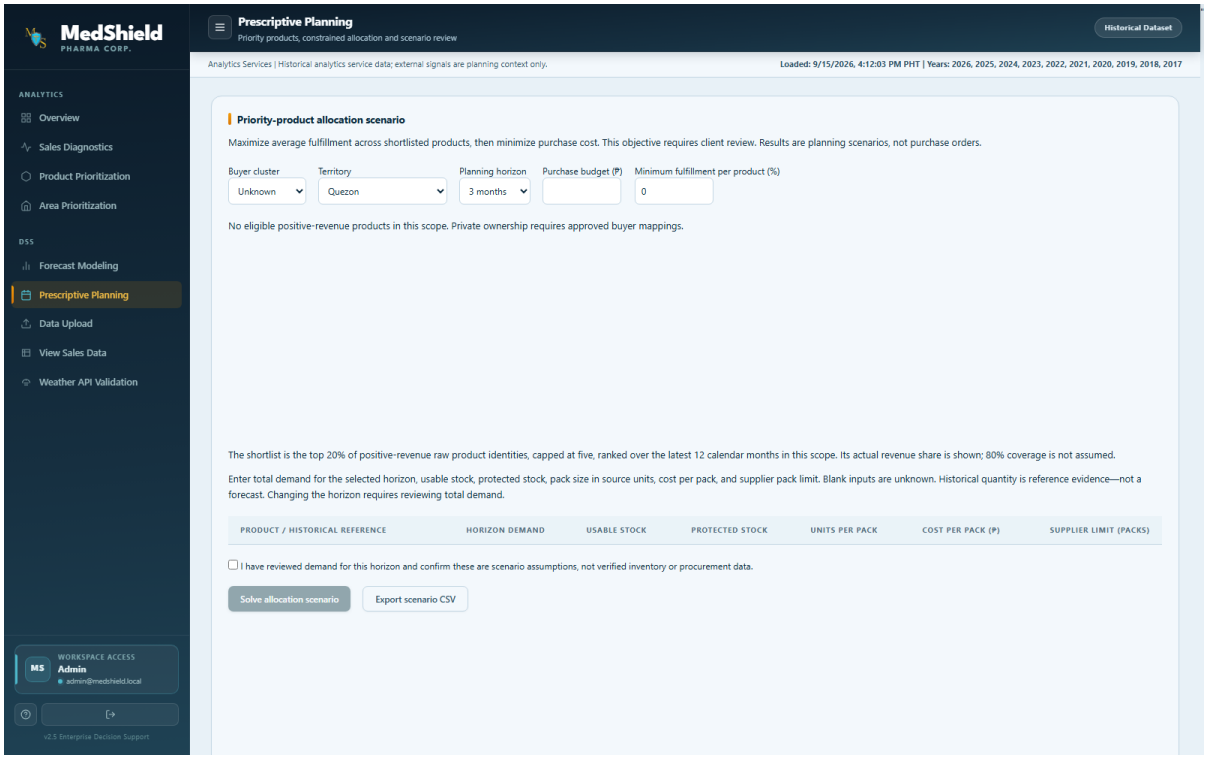


Figure 4.8 Prescriptive planning interface

Source: MedShield development interface capture dated 15 September 2026. Inputs and outputs are scenario assumptions, not purchase orders.

No actual wastage rate can be calculated from the present study because no expiry or write-off outcome dataset is available. The proposal's target of expiry-driven wastage at or below 5 percent remains a performance target for a future operational evaluation. Claiming that the system achieved the target would confuse a design objective with a measured result.

4.3 Dashboard

4.3.1 Implementation and Validation

The fifth objective is the most visible implementation result. The dashboard provides Sales Diagnostics, Product Prioritization, Area Prioritization, Forecast Modeling, and Prescriptive Planning, together with overview, transaction evidence, data upload, and weather validation views. The system supports dynamic annual and monthly filtering and keeps historical gaps visible. Its labels distinguish actual, candidate, proxy, baseline, blocked, and scenario evidence so the user can see whether a value is observed or still under review.

Table 4.12 Status of the five proposed modules

Module	Implemented result	Current boundary
Sales Diagnostics	Annual and monthly sales, margin, matched-period growth, quantity, heatmap, and STL evidence	Candidate financial definitions; missing months preserved
Product Prioritization	Positive-sales ranking and deterministic Pareto ABC	Source labels are not approved SKUs; XGBoost not published
Area Prioritization	Derived buyer-sector, geographic, and legacy-area views	Mapping ownership and mixed legacy labels remain under review
Forecast Modeling	Sales-only baselines, common-holdout metrics, candidate experiment evidence	No approved champion, intervals, or external regression
Prescriptive Planning	Constrained allocation scenarios with explicit assumptions and infeasibility handling	No real inventory, expiry, or purchase execution

Source: MedShield Current State, Analytics, and System Documentation.

The latest descriptive verification reported 70 of 70 publication checks passing for the version-16 candidate. This means the technical pipeline reconciled at the time of the review. It does not mean that the client approved the product master, area mappings, financial definitions, or model release. The QA record also reports 21 backend unit tests passing, a final eight-case descriptive browser run, five area and buyer-scope regressions, and six matched-period and growth cases with four explicitly retired legacy tests skipped. These sets are reported separately because they overlap in scope and should not be added into one inflated test count.

Table 4.13 Validation and acceptance evidence

Validation area	Observed result	Interpretation
Publication readiness	70 pass 0 fail 0 unverified	Technical candidate checks passed; business approval still open
Descriptive population	37,728 facts 108 calendar months 98 observed months	Counts and periods reconcile; ten months remain unavailable

Validation area	Observed result	Interpretation
STL	72 reconciled months and four components	Qualifying 2020 to 2025 segment only
Backend unit tests	21 passed	Covers current backend logic including filter validation
Final descriptive browser run	8 passed	Live version-bound descriptive behaviors and exports
Area and buyer regression	5 passed	Scoped classification behavior
Matched-period and growth suite	6 passed 4 legacy skipped	Skipped tests are retired and are not counted as passes
Predictive warehouse candidate	1,092 rows and 15 evaluations persisted	Candidate experiment evidence; active model pointer remains empty

Source: MedShield System QA Review, Current State, Database, and Predictive Operations records dated 27 September 2026.

4.4 Summary of Findings

4.4.1 Historical Patterns

The authorized candidate shows ₱626.95 million in observed net sales across 2017 to 2025. Recorded activity is uneven by year and by month, with ten missing months in the early history and a large 2025 value that requires careful provenance interpretation. May and June have the highest observed calendar-month averages. STL provides a separate 2020 to 2025 trend, seasonal, and residual decomposition without imputing the earlier gaps.

4.4.2 Forecast Comparison

On the comparable fixed-origin Government holdout, last observed value performed better than seasonal naive, but its WAPE remained 62.71 percent. In the larger offline experiment, last value also remained selected, with 83.20 percent development WAPE and 37.76 percent retrospective WAPE. These errors, the lack of uncertainty support, and the absence of approved external joins justify the decision to keep the model unpublished.

4.4.3 Product and Area Priorities

SPEEDA 2.5IU/0.5ML is the highest-revenue source product label in the current candidate, while GOVERNMENT is the highest-revenue derived area label. Both findings are useful for review, but neither should be treated as a final canonical SKU or a geographic market-share statement. The approved product master is empty, and the legacy area field mixes different commercial meanings.

4.4.4 Planning Scenarios

The system can produce constrained allocation scenarios after a user supplies operational assumptions. The resulting plan identifies feasible allocations under those assumptions and exposes infeasible constraints. It cannot determine real EOQ, ROP, safety stock, or expiry reduction until the client publishes the required inventory and procurement data.

4.4.5 Decision Support Value

MedShield improves demand visibility by placing source coverage, KPI definitions, historical patterns, priority candidates, forecast errors, and planning assumptions in one review path. Its current contribution is transparency and structured comparison. The study does not establish usability through participant ratings because no REC-approved user evaluation evidence is included in the available record. System tests demonstrate behavior; they do not substitute for user acceptance or perceived usefulness data.

4.4.6 Discussion of the North Star

The evidence supports the first half of the North Star more strongly than the second. MedShield improves demand visibility because it consolidates historical sales, preserves data gaps, shows the basis of each KPI, and makes the status of models and mappings visible. It also creates a path from historical priority to a reviewed allocation scenario. These are practical decision-support gains because they reduce the amount of hidden judgment in a planning discussion.

The study does not yet demonstrate reduced expiry-driven losses. That outcome requires a baseline of expiry or write-off losses, comparable post-implementation observations, and operational inventory inputs. The current system can help prepare such an evaluation by recording the assumptions and decisions used in each scenario. Until outcome data exist, the defensible conclusion is that MedShield provides the information and controls needed to support planning; it has not proven the financial effect of those plans.

The findings also show why simple models and explicit release gates are valuable. A more complex forecasting method did not earn promotion merely by using Prophet or external regressors. The system retained last value because it performed best among the tested candidates, then withheld publication because the errors and uncertainty evidence were still weak. This is

consistent with the role of a decision-support system: it should make the basis of a recommendation inspectable and leave the final action to an authorized user.

4.4.7 Limitations

- The Gold sales objects remain candidate publications pending named business and finance approval.
- The product master has no approved entries, so product analysis uses source labels and cannot standardize aliases or pack conversions.
- The legacy area field combines geographic, customer-type, and business-line labels; final mapping approval is still required.
- Ten months have no observed transactions. The study preserves these gaps, but it cannot independently prove whether they represent missing files or no business activity.
- DOH and PAGASA rows are not approved for territory-level forecasting, while NASA POWER remains a separately labeled proxy.
- No predictive model or uncertainty interval is approved for operational publication.
- Inventory, expiry, supplier, lead-time, service-level, ordering-cost, holding-cost, and stock-position data are unavailable.
- The study contains technical and browser validation but no completed REC-approved participant evaluation or final client acceptance record.

4.4.8 Overall Synthesis

MedShield now has a working and testable historical decision-support foundation. The system reconciles 37,728 candidate sales facts, presents 2017 to 2025 performance, preserves ten missing months, decomposes a qualifying 72-month series through STL, ranks product and area labels, compares common-holdout baselines, and exposes constrained planning scenarios through an authenticated dashboard architecture. These results satisfy the descriptive and presentation objectives most directly.

The chapter also records where the project has not yet reached its intended outcome. Product identity and area mappings still need owner approval. External signals are not forecast-eligible. The selected forecast remains unpublished because its errors are high and uncertainty support is incomplete. Prescriptive outputs remain scenarios, and the expiry-wastage target has

not been measured. Presented this way, the system is defendable as a historical DSS that improves visibility and review discipline while keeping operational claims within the evidence.

References

Carvajal, T. M., Viacrusis, K. M., Hernandez, L. F. T., Ho, H. T., Amalin, D. M., and Watanabe, K. (2018). Machine learning methods reveal the temporal pattern of dengue incidence using meteorological factors in metropolitan Manila, Philippines. BMC Infectious Diseases, 18, 183.

Hyndman, R. J., and Koehler, A. B. (2006). Another look at measures of forecast accuracy. International Journal of Forecasting, 22(4), 679–688.

MedShield Capstone Group 9. (2026). Capstone project proposal objectives and conceptual framework. Unpublished manuscript.

MedShield Capstone Group 9. (2026). Analytics. Internal project documentation, reviewed 27 September 2026.

MedShield Capstone Group 9. (2026). Architecture. Internal project documentation, reviewed 27 September 2026.

MedShield Capstone Group 9. (2026). Current State. Internal project documentation, reviewed 27 September 2026.

MedShield Capstone Group 9. (2026). Descriptive calculation and STL receipts. Internal validation evidence.

MedShield Capstone Group 9. (2026). Predictive Operations. Internal model-evaluation documentation, reviewed 27 September 2026.

MedShield Capstone Group 9. (2026). System QA Review. Internal verification record, reviewed 27 September 2026.

MedShield Capstone Group 9. (2026). System Documentation. Internal technical and operational overview.